Status	Finished
Started	Friday, 16 October 2026, 8:11 PM
Completed	Friday, 16 October 2026, 8:14 PM
Duration	2 mins 56 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Sorting
- b. Searching
- c. Rearranging
- d. Merging

The correct answer is: Sorting

Question **2**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It only works with integer arrays
- b. It sorts data out of the box efficiently
- c. It requires external libraries
- d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **3**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It only works on small datasets
b. It avoids the need for recursion
c. It allows for efficient parallel processing and sorting of data
d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **4**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Being the most efficient sorting algorithm
b. Using the divide-and-conquer approach
c. Sorting data in a non-sequential manner
d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Bubble Sort
b. Insertion Sort
c. Quick Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **6**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. All of the above
- b. Duplicates
- c. Searching
- d. Selection

The correct answer is: All of the above

Question **7**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Deleting data from a list
- b. Arranging data in a particular format
- c. Searching for data in a list
- d. Inserting data into a list

The correct answer is: Arranging data in a particular format

Question **8**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Selection
- b. Insertion
- c. Distribution
- d. Extraction

The correct answer is:
Insertion

Question **9**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By comparing elements sequentially
b. By using the bubble-up method
c. By sorting data in a single pass
d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **10**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It uses the heap data structure
b. It bubbles up the largest element in each pass
c. It recursively sorts subproblems
d. It divides the list into sublists

The correct answer is: It bubbles up the largest element in each pass

Question **11**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It complicates the analysis process
b. It makes the analysis slower
c. It has no effect on the analysis process
d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Comparing each element with the others
- c. Combining sorted sublists
- d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **13**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort is less efficient for large lists
- b. Bubble Sort does not guarantee sorted order
- c. Bubble Sort is difficult to implement
- d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **14**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Bubble Sort
- b. Binary Search
- c. Merge Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging

b. Sorting

c. Complexity

d. Searching

The correct answer is:
Complexity